



Milestones for Placement Portal Application V2 - Modern Application Development II

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Milestones for Placement Portal Application – V2

Modern Application Development II

Project: Placement Portal Application V2 (MAD-II)

- The project is divided into **0th milestone, Core Requirements (8 Milestones), and Recommended Functionalities & Optional Enhancements (2 Milestones)**.
 - **All core milestones can be completed in any order.**
 - Recommended and optional enhancements should only be started after the core milestones are done.
 - Each milestone includes a Git commit message. When completed, **use a unique commit message and push your updates to GitHub.**
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Completion Progress: 5%

- Create a **private GitHub repository**.
- Add **README.md** with a short project description.
- Add **.gitignore** for Python, Flask, Vue, SQLite, Redis cache files. (not compulsory).
- Commit and push initial setup to the repository.
- Refer to the [git tracker document](#) to add **MADII-cs2006** as your collaborator, so that we can track your project. (**Requests will be accepted after term starts.**)
- Github Adding Collaborator Video: [Adding Github Collaborator using VSCode and WSL](#)
- **This milestone needs to be completed by 14th July 2026.**

Git Commit Message: Milestone-0 PPA-V2 Setup



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✓ Expected Time: 5–7 days
📊 Completion Progress: 15%

- Create models for:
 - User (role: Admin / Company / Student)
 - Company (profile, industry, location, etc.)
 - Student (profile, education, skills, resume, etc.)
 - Job Position (company_id, title, salary, skills_required, etc.)
 - Application (student_id, job_id, status, date, etc.)
 - Placement (student_id, company_id, position, salary, joining_date, etc.)
 - Any other if you need.
- Define relationships (1:n, n:n where required, e.g., Company–Job Position, Student–Application, Application–Placement).
- Ensure Admin user is **pre-created programmatically** through a Python script/file (using SQLite query or SQLAlchemy ORM).

Git Commit Message: USE UNIQUE COMMIT MESSAGE ACCORDING



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(Admin/Company/ Student) (Flask+Vue)



Expected Time: 5 days



Completion Progress: 10%

- Implement authentication using **Flask-Security or JWT-based tokens**.
- Students: self-register + login.
- Companies: register + login only (approved by Admin).
- Admin: predefined login only (no registration).
- Redirect users to **role-specific dashboards** after login (Admin, Company, Student).

Git Commit Message: USE UNIQUE COMMIT MESSAGE ACCORDING TO MILESTONE

Milestone: Admin Dashboard and Management (Flask+Vue)



Expected Time: 7–9 days



Completion Progress: 15%



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- Approve and Remove Company profiles.
- Approve and Remove job posting/placement drives created by Companies.
- Search companies (by name / industry).
- Search students (by name / ID / contact).
- View and manage all job postings and applications.
- Blacklist/Deactivate companies and students from the system.

Git Commit Message: USE UNIQUE COMMIT MESSAGE ACCORDING TO MILESTONE

Milestone: Company Dashboard and Job/Application Management (Flask+Vue)



Expected Time: 7 days



Completion Progress: 15%

Company functionalities:

- Companies will register their profile.



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- Post new job positions with required skills, experience, salary, and benefits.
- View list of applicants for each job.
- Shortlist or reject applicants with feedback.
- Manage job posting status (Active / Closed).
- Schedule interviews with shortlisted candidates.
- Share acceptance or rejection status (Shortlisted / Selected / Rejected) to applicants.

Git Commit Message: USE UNIQUE COMMIT MESSAGE ACCORDING TO MILESTONE

Milestone: Student Dashboard and Job Application System (Flask+Vue)



Expected Time: 7 days



Completion Progress: 13%

Student functionalities:



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- Apply for jobs and track application status.
- View applied jobs with detailed application status.
- View interview schedules and feedback from companies.
- Download offer letters or placement confirmations.

Git Commit Message: USE UNIQUE COMMIT MESSAGE ACCORDING TO MILESTONE

Milestone: Job Application History and Status Tracking (Flask+Vue)

✓ Expected Time: 5–6 days
📊 Completion Progress: 12%

- Store and display complete application and placement history.
- Prevent duplicate applications for the same job posting.
- Ensure only approved companies can create placement drives.
- Ensure students can view and apply only to approved placement



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- Admin and Company can view student profiles and applications, Students can view their own records.

Git Commit Message: USE UNIQUE COMMIT MESSAGE ACCORDING TO MILESTONE

Milestone: Backend Jobs – Interview Reminders, Placement Reports, and Triggered Jobs (CSV Export using Celery + Redis) (Flask)

✓ Expected Time: 7–8 days
📊 Completion Progress: 10%

- **Setup Celery workers, Celery Beat, and Redis server.**
- **Interview Reminder Job:** send reminders (Email / GChat / SMS) to students with scheduled interviews.
- **Placement Report Job:** generate monthly reports for companies (HTML / PDF) with application statistics, placements, and analytics.
- **User-triggered CSV Export:** students and companies can export application and placement history asynchronously; alert sent once the batch job is complete.

Git Commit Message: USE UNIQUE COMMIT MESSAGE ACCORDING



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Milestone: API Performance Optimization and Caching Using Redis (Flask)



Expected Time: 3 days



Completion Progress: 5%

- Use **Redis caching** for API optimization.
- Cache frequently used endpoints (job listings, company search, student search).
- Implement proper cache expiry and refresh policies.

Git Commit Message: USE UNIQUE COMMIT MESSAGE ACCORDING TO MILESTONE

Recommended / Optional Enhancements (2 Milestones)

These milestones are recommended but not mandatory.



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Improve **UI styling** using Bootstrap & VueJS.

- Make the web app mobile-responsive.
- Implement **frontend validation** (JavaScript) and **backend validation** (Flask).
- Add "Add to Home Screen" (PWA like behavior).
- Consistent design across devices.

Git Commit Message: USE UNIQUE COMMIT MESSAGE ACCORDING TO MILESTONE

Milestone: Reports, Charts and ATS (Vue)

✓ Expected Time: 10 days

- Integrate Chart.js for analytics (placement trends, job demand by skills, application funnels).
- **Public Landing Dashboard Page (pre-login)** showing **aggregated monthly placement statistics** (read-only, no sensitive data).



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Final Submission

Milestone: Final Project Submission



Expected Time: 2 days



After Completion Progress: 100%

- Submit **ZIP with code + 3–5 page report**.
- Submit a 3-10 minute demo video (Google Drive with access to **anyone with the link**).
- Include a **video link in the report**.
- Include a **declaration of AI usage %** in the same pdf report (**if used or not declaration is mandatory**).
- Use this video [How to Submit App Dev 1 & 2 Projects | Full Submission Guide + Viva Workflow | IITM BS](#) on how to submit the app in the portal.

Git Commit Message: Milestone-PPA-V2 Final-Submission



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- 1. **Use Git commits** for each milestone.
- 2. **Test functionality** after completing each milestone.
- 3. **Log issues** encountered and their resolutions in a **README.md** or GitHub Issues section.

Timeline Overview

Project Completion Overview

Milestone	Average Time Estimate	Split (%)	Progress (%)
GitHub Setup	1 day	5%	5%
DB Models & Schema	5-7 days	15%	20%
Auth & RBAC	5 days	10%	30%
Admin Dashboard	7-9 days	15%	45%
Company Dashboard	7 days	15%	60%
Student Dashboard	7 days	13%	73%
Placement Status	5-6 days	12%	85%
Celery Jobs	7-8 days	10%	95%
Redis Caching	3 days	5%	100%



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Milestone	Time
Responsive UI/PWA	6 days
Reports, Charts & ATS	10 days

Final Submission Overview

Milestone	Average Time Estimate
Final Project Submission Details	2 days

Note:

- Time estimates are averages. Spend time, as needed, to understand and complete each milestone. Completing Milestone 0 and 8 core requirements are taken for grading in Viva1. After completing the core requirements, there are 2 additional milestones for enhancements. These enhancements are not graded but can be completed for improved functionality or to be considered for the Best Course Project nomination.*
- Students can complete the core milestones in any order as per their approach to doing the project. They need not follow the same sequence as they appear in this document.*
- Students can have any number of commits in their project repository, but for a milestone to be considered as "complete", the student must make a commit with a unique commit message of that milestone.*
- It is mandatory to push the entire codebase to the GitHub before you submit the code on the viva-portal.*



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